Reconstructing Tumor Evolutionary Histories and Clone Trees in Polynomial-time with SubMARine

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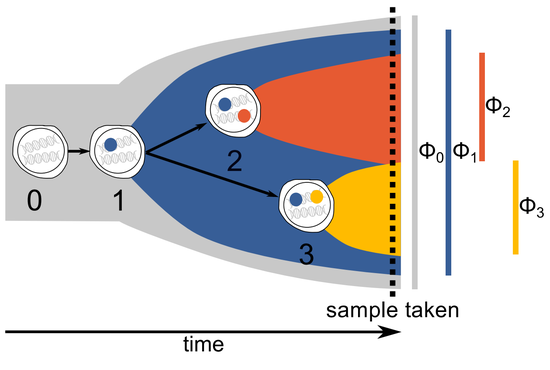
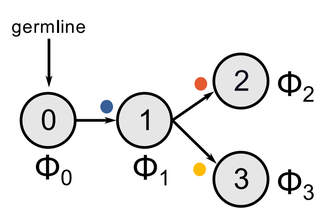
Inferring clone trees

Tumors evolve by accumulating mutations over time, leading to a heterogeneous set of genetically distinct cancer cells. Knowledge about the evolutionary history can help us understanding how tumors develop and respond to treatment. Given the sequencing data of a tumor sample, different methods exist to infer subclonal frequencies and build clone trees from them.



What is a clone tree?

A clone tree is a directed, rooted tree whose nodes correspond to subclones and whose edges link parental subclones to their direct descendants. A subclone is a group of cells descending from same founder cell; and corresponds to a subtree (or clade) in the phylogeny of the cancerous subpopulations. Each subclone k is associated with set of mutations and its subclonal frequency $\phi(k,n)$ for each sample n. The root of the clone tree is called the germline and represents the embryonic stem cell, which is the founder of all cancer cells (and all other cells in the body).



When constructing a clone tree from a matrix of subclonal frequencies, there exist two key challenges:

1. Scaling algorithms to a large number of subclones.
2. Efficiently representing uncertainty in the reconstruction.

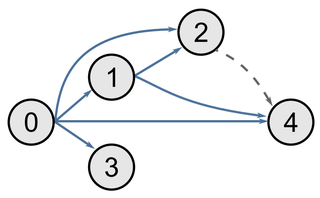
In order to address these challenges, we have developed SubMARine, a polynomial-time algorithm that builds partial clone trees.

What is a partial clone tree?

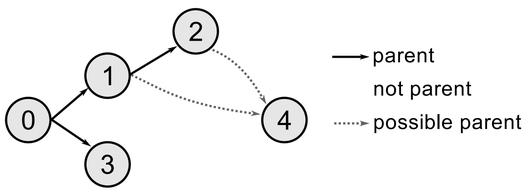
A partial clone tree defines a subset of the pairwise ancestral relationships between the subclones. We represent it in an ancestry matrix Z:

$$Z(k,k') = \begin{cases} 1 & \text{if subclone } k \text{ is an ancestor of subclone } k' \\ 0 & \text{if subclone } k \text{ is not an ancestor of subclone } k' \\ -1 & \text{if subclone } k \text{ could be an ancestor of subclone } k' \\ & \text{(aka undefined)} \end{cases}$$

k \ k'	0	1	2	3	4
0	0	1	1	1	1
1	0	0	1	0	1
2	0	0	0	0	-1
3	0	0	0	0	0
4	0	0	0	0	0



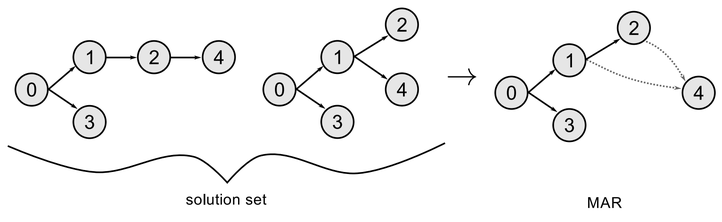
A more intuitive visualization of this partial clone tree is this possible parent DAG:



A partial clone tree is not a tree itself but represents a set of clone trees. These are the clone trees consistent with the ancestral relationships defined in the partial clone tree that select their parents from the possible parent set.

Maximally-constrained ancestral reconstruction (MAR)

Given a subclonal frequency matrix ϕ, for which a set of valid clone trees exists, the maximally-constrained ancestral reconstruction (MAR) is a special partial clone tree that efficiently represents the uncertainty in the set of clone trees. It defines the maximal set of all of the ancestral relationships shared by the solution set of clone trees and leaves undefined all relationships that vary.



Given a set of valid clone trees, it is trivial to build their MAR. However, usually this set is not given.

SubMARine: Approximating the MAR

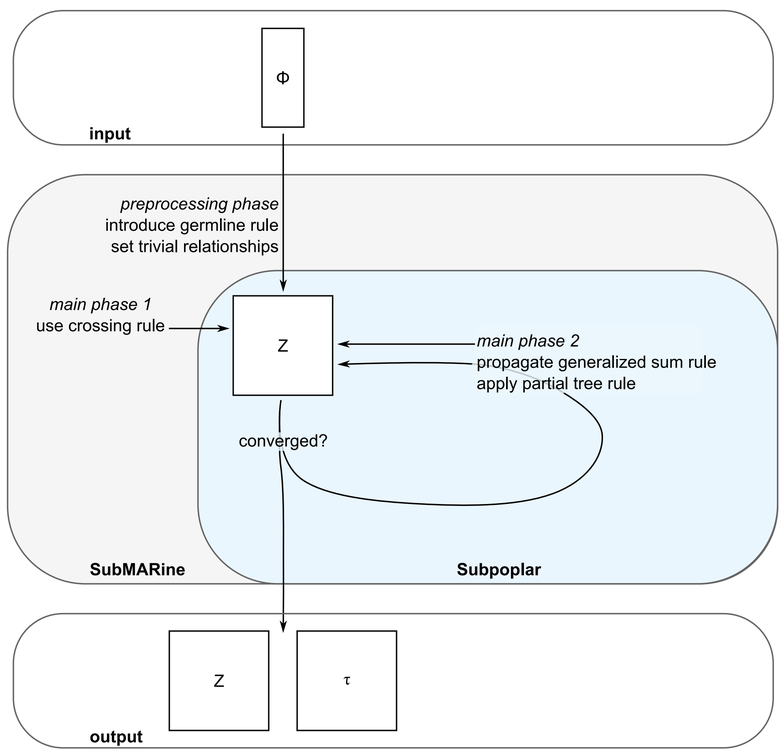
Given a subclonal frequency matrix ϕ, our polynomial-time algorithm SubMARine approximates the MAR by constructing the subMAR. This is another partial clone tree with three important properties:

1. The subMAR is unique,
2. its defined ancestral relationships are a subset of those in the MAR,
3. and all valid clone trees of ϕ are completions of it.

The basic SubMARine algorithm

In its basic version, SubMARine takes the subclonal frequency matrix ϕ as input and creates an ancestry matrix Z, whose relationships are initially all undefined. SubMARine defines relationships by propagating different validity rules until convergence and then outputs the ancestry matrix together with the matrix τ, which contains the possible parents for each subclone.

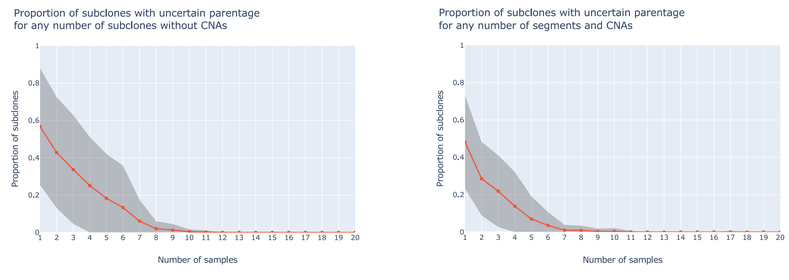
The **germline rule** ensures that the germline is ancestor of all subclones. The **partial tree rule** guarantees that the partial clone tree has tree properties, by ensuring that each subclone cannot have more than one definite parent and that the partial clone tree is transitive. The **generalized sum rule** assures that the frequency of a subclone is at least as large as the frequency sum of its definite children. By sorting the subclones in descending order of their average subclonal frequency, we can derive *trivial relationships* (lower left triangle) in the ancestry matrix. The *crossing rule* is another consequence of the generalized sum rule; it states that two subclones are not ancestors of one another if one subclone has a higher frequency than the other in one sample but a lower one in an another sample. Also our *Subpoplar* algorithm takes care of propagating the generalized sum rule, by not making a subclone a child of another subclone if the rule was violated but making a subclone a definite child of another subclone if it is its only possible parent.



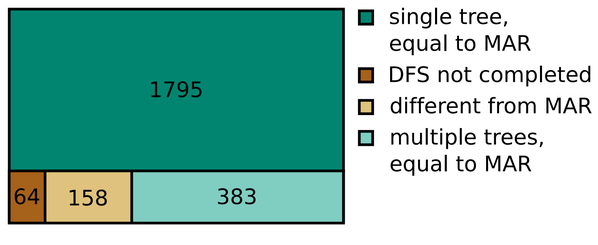
Next to the basic version of SubMARine, we also developed an extended version that takes subclonal copy number aberrations into account. Furthermore, we allow SubMARine to deal with noisy subclonal frequencies by introducing a noise buffer.

Results on simulated data

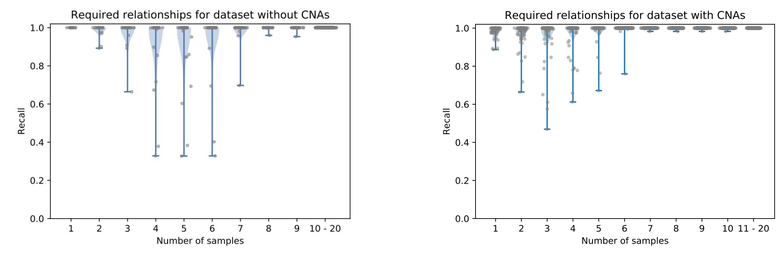
We applied SubMARine to the noise-free simulated subclonal frequencies of 600 clones trees without and 1800 clone trees with subclonal CNAs, with 5, 20 and 50 subclones. SubMARine constructed each subMAR in less than 70 seconds on a single thread with less than 1 GB of memory. We observed that the more the input data was constrained, either by a higher number of samples or by containing CNAs, the less subclones had uncertain parentage, hence the less undefined relationships were present in the subMAR.



About 3/4 of constructed subMARs contained no undefined relationships, thus describe a single tree and equal their MAR. To compare the other subMARs to their MAR, we implemented a depth-first search (DFS) that enumerates all valid completing clone trees.

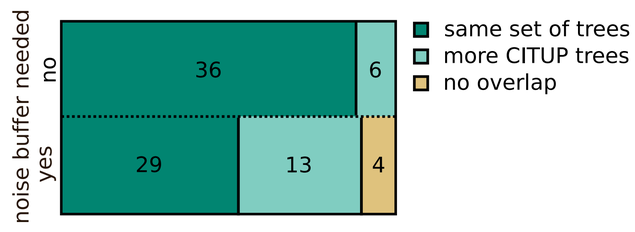


In order to evaluate how well the subMARs matched their MARs, we computed the recall of defined relationships. We observed that the more the input data was constrained, the more the subMAR resembled the MAR.



Results on TRACERx data

We also applied SubMARine to the noisy subclonal frequencies of 88 patients of the large, mutli-sample lung cancer TRACERx study (Jamal-Hanjani et al., The New England Journal of Medicine, 2017). SubMARine constructed each subMAR in less than 40 seconds. In contrast to that, for three patients with more than 10 subclones, the authors of the study could not build clone trees with CITUP (Malikic et al., Oxford Bioinformatics, 2015) and had to construct them manually. The subMARs of all patients equalled their MARs. We compared the subMAR-completing clone trees with the trees reported in the paper and found that the results were largely identical.



Summary

We introduced the notion of a partial clone tree. Our polynomial-time method SubMARine scales to a large number of subclones and efficiently represents uncertainty in the subMAR. The more constrained the input data is, the more the subMAR resembles the MAR.

Acknowledgments

LKS was partially funded by GRK 1906/1, and is now funded by a MITACS elevate postdoctoral fellowship. JW is funded by an NSERC scholarship. GR is partly funded by the "Swiss Molecular Pathology Breakthrough Platform", funded by the ETH Special Focus Area "Personalized Health Related Technologies", grant number #106. QM is supported by an Associate Investigator Award from the Ontario Institute of Cancer Research (which partially supports LKS), a subgrant from the Canadian Centre for Computational Genomics genomics technology platform funded by Genome Canada, and holds a Canada CIFAR AI chair.

https://github.com/morrislab/submarine
https://www.biorxiv.org/content/10.1101/2020.06.11.146100v1